

On the Incidence Ranking of Paths, Cycles and Grids*

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Abstract

Let G be a graph with vertex set $V(G)$ and edge set $E(G)$, respectively. We denote by $I(G) = \{(v, e) : v \in V(G), e \in E(G), \text{ and } v \text{ is an end-vertex of } e\}$ the set of incidences of G and investigate the following graph labeling problem. An incidence ranking of a graph G is a mapping $f: I(G) \rightarrow \mathbb{Z}^+$ such that for any path between two distinct incidences $i, j \in I(G)$ with $f(i) = f(j)$, there exists an incidence k in the path with $f(k) > f(i)$. An incidence ranking f is a k -incidence ranking, for some $k \in \mathbb{Z}^+$, if $k = \max\{f(i) : i \in I(G)\}$. The incidence ranking number of G , denoted $\chi_{ir}(G)$, is defined by $\chi_{ir}(G) = \min\{k \in \mathbb{Z}^+ : f \text{ is a } k\text{-incidence ranking of } G\}$. The incidence ranking problem is to determine the incidence ranking number of a given graph. In this paper, we obtain the incidence ranking numbers of paths, cycles and grids $P_m \times P_n$ for $m = 2, 3$ and $n \geq 2$ arbitrary.

Keyword: graph labeling; graph coloring; incidence ranking; grid graphs;

1 Introduction

Given a graph $G = (V, E)$, a graph labeling problem usually demands to provide with a function that maps the vertices of $V(G)$, edges of $E(G)$ or other elements of G to a subset of integers and is subject to certain constraints. For instance, a *vertex coloring* of a graph G is a vertex labeling in which no two adjacent vertices have the same label. Also, a *vertex ranking* (or simply, *ranking*) of a graph G is a mapping f from $V(G)$ to a set of positive integers such that, for any two distinct vertices $u, v \in V(G)$ with $f(u) = f(v)$, every path between

u and v contains some vertex w with $f(w) > f(u)$. The *ranking number* of G , denoted $\chi_r(G)$, is the smallest number of integers required in a ranking of G . Obviously, ranking is a vertex coloring with an additional condition imposed. Note that if we apply the notion of a graph labeling that uses edges instead of vertices in the condition, the problems of *edge coloring* and *edge ranking* can be defined similarly.

The concept of ranking number was first defined for tree only. Iyer et al. [17] gave an $\mathcal{O}(n \log n)$ algorithm for finding an optimal ranking of an n -vertex tree, and Schäffer [26] further made an improvement in $\mathcal{O}(n)$ time. The study of ranking problem has interesting applications on the assembly of products in manufacturing systems and the design of VLSI layout [24, 27]. It has already known that the ranking problem is NP-hard for general graphs [22] and even when it is restricted to bipartite graphs [2]. More results about the ranking number of graphs had been obtained, see e.g. [3, 7, 9, 10, 12, 13, 15, 19, 23]. In particular, [1, 5, 6] and [25] are related to grid graphs. By contrast, for edge ranking of graphs, we refer to [8, 18, 20, 21].

For a graph G , a vertex $v \in V(G)$ and an edge $e \in E(G)$ are said to be *incident* with each other (i.e., v is incident with e , or e is incident to v) if v is an end-vertex of e . Accordingly, the set of *incidences* of G is defined by $I(G) = \{(v, e) : v \in V(G), e \in E(G), \text{ and } v \text{ is incident with } e\}$. Two incidences (v, e) and (w, f) are *adjacent* if one of the following conditions holds: (i) $v = w$, (ii) $e = f$ or (iii) the edge vw is equal to e or f . Otherwise, (v, e) and (w, f) are *nonadjacent*. An *incidence coloring* of a graph G is a coloring of its incidences in which adjacent incidences are assigned different colors. The *incidence coloring number* of G , denoted $\chi_i(G)$, is the smallest number of colors in an incidence coloring of G .

Brualdi and Massey [4] first introduced the problem of finding the incidence coloring number of a

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graph G . So far the incidence coloring number are known for a few special graph classes such as trees, complete graphs [4], square meshes, hexagonal meshes, honeycomb meshes [16], Halin graphs with $\Delta(G) \geq 5$ and outerplanar graph with $\Delta(G) \geq 4$ [29], where $\Delta(G)$ is the maximum degree of G . Also, there are some results related to the lower bound and the upper bound of incidence coloring number of graphs, see e.g. [11, 14, 28] and their references.

In this paper, we define the following graph labeling problem. Let $\mathbb{Z}^+$ denote the set of positive integers. An *incidence ranking* of a graph G is a mapping $f: I(G) \rightarrow \mathbb{Z}^+$ such that for any path between two distinct incidences $i, j \in I(G)$ with $f(i) = f(j)$, there exists an incidence k in the path with $f(k) > f(i)$. An incidence ranking f is a k -*incidence ranking*, for some $k \in \mathbb{Z}^+$, if $k = \max\{f(i): i \in I(G)\}$. The *incidence ranking number* of G , denoted $\chi_{ir}(G)$, is defined by $\chi_{ir}(G) = \min\{k \in \mathbb{Z}^+: f \text{ is a } k\text{-incidence ranking of } G\}$. An incidence ranking f of a graph G is *optimal* if f is an $\chi_{ir}(G)$ -incidence ranking. The *incidence ranking problem* is to determine the incidence ranking number $\chi_{ir}(G)$ of a given graph.

Throughout the paper, P_n and C_n denote the path and cycle on n vertices, respectively. Also, the grid graph $P_m \times P_n$ is the Cartesian product of P_m and P_n . For notational convenience, we write $G_{m,n}$ instead of $P_m \times P_n$. In this paper, we obtain the incidence ranking numbers of paths, cycles and grids $G_{2,n}$ and $G_{3,n}$, respectively. In addition, we provide upper bounds of $\chi_{ir}(G_{m,n})$ for $m = 4, 5$ and $n \geq 2$ arbitrary.

2 Preliminary

In this section, we give some elementary results related to incidence ranking.

Lemma 1. *For any incidence ranking of a connected graph G , the largest label must be unique.*

Proof. Suppose that there exist two incidences $i, j \in I(G)$ which have the largest label k . Then any path between i and j will not contain an incidence with label larger than k . This is a contradiction. $\square$

Lemma 2. *Let H be a subgraph of a graph G . Then $\chi_{ir}(H) \leq \chi_{ir}(G)$.*

Proof. Suppose $\chi_{ir}(H) > \chi_{ir}(G)$. Then we could relabel the incidences of H using the corresponding labels used in the incidence ranking of G . This will produce an incidence ranking with fewer labels, and hence a contradiction. $\square$

Let $\mathbb{Z}_n^+ = \{1, 2, \dots, n\}$. Given k graphs $H_1, H_2, \dots, H_k$, the *union* of these k graphs, denoted $\bigcup_{i=1}^k H_i$, is a graph G with $V(G) = \bigcup_{i=1}^k V(H_i)$ and $E(G) = \bigcup_{i=1}^k E(H_i)$.

Lemma 3. *If $G = \bigcup_{i=1}^k H_i$, where $H_1, H_2, \dots, H_k$ are pairwise vertex-disjoint, then $\chi_{ir}(G) = \max\{\chi_{ir}(H_i): i \in \mathbb{Z}_k^+\}$.*

Proof. By Lemma 2, $\chi_{ir}(G) \geq \max\{\chi_{ir}(H_i): i \in \mathbb{Z}_k^+\}$. On the other hand, for each $j \in \mathbb{Z}_k^+$, we suppose that f_j is an optimal incidence ranking of H_j . Define a labeling f of G as $f(i) = f_j(i)$ for $i \in I(H_j)$. Then, clearly, f is an incidence ranking of G . Thus, $\chi_{ir}(G) \leq \max\{\chi_{ir}(H_i): i \in \mathbb{Z}_k^+\}$. $\square$

In what follows, we consider the relation between the incidence ranking numbers and the minimal edge cut sets. For a graph G , a subset $S \subset E(G)$ is called an *edge-cut* of G if the removal of S , denoted $G - S$, results in a disconnected graph. Moreover, an edge-cut S is *minimal* if any proper subset of S is not an edge-cut of G . Obviously, if S is a minimal edge-cut of G , then $G - S$ contains only two connected components. For an edge-cut S of G , we use the notation $\chi_{ir}^*(G|S)$ to denote the number $|S| + \chi_{ir}(G - S) = |S| + \max\{\chi_{ir}(H): H \text{ is a component of } G - S\}$. The following lemma is a direct consequence of the definition of incidence ranking.

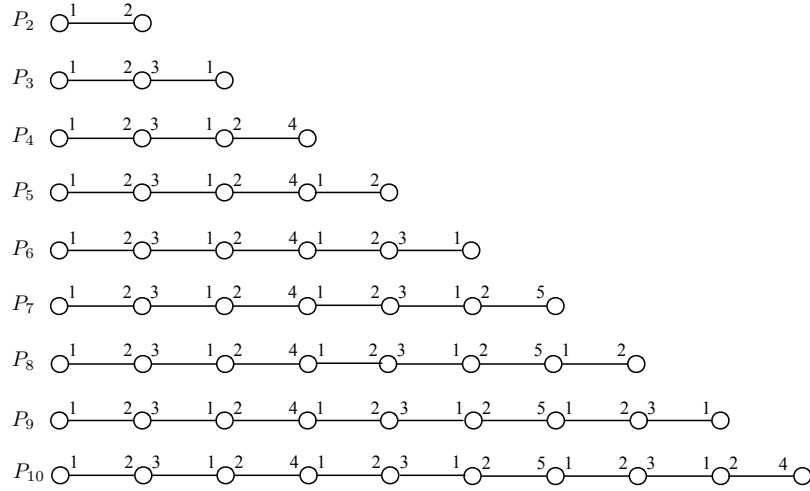
Lemma 4. *Let f be an k -incidence ranking of a graph G . If $m = \max\{\ell \in \mathbb{Z}_k^+: |f^{-1}(\ell)| \geq 2\}$ and $\hat{I}(G) = \{i \in I(G): f(i) \geq m + 1\}$, then $\{e \in E(G): (v, e) \in \hat{I}(G) \text{ for } v \in V(G)\}$ is an edge-cut of G .*

Lemma 5. *For any graph G , $\chi_{ir}(G) \geq \min\{\chi_{ir}^*(G|S): S \text{ is a minimal edge-cut of } G\}$.*

Proof. We show that there exists a minimal edge-cut $S \subseteq E(G)$ such that $\chi_{ir}(G) \geq \chi_{ir}^*(G_S)$. Consider an optimal incidence ranking f of G . Let $m = \max\{\ell \in \mathbb{Z}_k^+: |f^{-1}(\ell)| \geq 2\}$ and $\hat{I}(G) = \{i \in I(G): f(i) \geq m + 1\}$. By Lemma 4, the set $S' = \{e \in E(G): (v, e) \in \hat{I}(G)\}$ is an edge-cut of G . Let $S \subseteq S'$ be a minimal edge-cut of G and assume that $S' - S = \{e_1, e_2, \dots, e_t\}$. Define a labeling f' of $I(G - S)$ as follows:

$$f'(v, e) = \begin{cases} f(v, e) & \text{if } (v, e) \notin \hat{I}(G); \\ m + i & \text{if } (v, e) \in \hat{I}(G) \text{ and } e = e_i \text{ for } i \in \mathbb{Z}_t^+. \end{cases}$$

Clearly, f' is an incidence ranking of $G - S$. Since f is an optimal incidence ranking of G , we have $\chi_{ir}(G - S) \leq m + |S' - S| = \chi_{ir}(G) - |S|$. Hence, $\chi_{ir}(G) \geq |S| + \chi_{ir}(G - S) = \chi_{ir}^*(G|S)$. $\square$

Figure 1: Incidence ranking of P_n for $n \leq 10$.

3 $\chi_{ir}(P_n)$ and $\chi_{ir}(C_n)$

In this section, we study the problem of incidence ranking on paths and cycles. We begin by showing $\chi_{ir}(P_n)$ for $n \leq 10$ in Figure 1.

Let $P_n = v_0, e_1, v_1, e_2, v_2, \dots, e_{n-1}, v_{n-1}$ denote a path on n vertices, where vertices and edges are appeared alternately. Clearly, for any edge $e_i \in E(P_n)$, it associates with two incidences (v_{i-1}, e_i) and (v_i, e_i) . To acquire an optimal ranking, we let $f: I(P_n) \rightarrow \mathbb{Z}^+$ be a functions defined as follows: for $1 \leq i \leq n-1$,

$$f(v_{i-1}, e_i) = \begin{cases} 3 & \text{if } i \equiv 2 \pmod{3} \\ 1 & \text{if } i \equiv 1 \pmod{3} \\ 2 & \text{if } i \equiv 0 \pmod{3} \end{cases} \quad (1)$$

and

$$f(v_i, e_i) = \begin{cases} 1 & \text{if } i \equiv 2 \pmod{3} \\ 2 & \text{if } i \equiv 1 \pmod{3} \\ 4 & \text{if } i \equiv 0 \pmod{3} \text{ and } i \equiv 1 \pmod{2} \\ f(v_{\frac{i}{2}}, e_{\frac{i}{2}}) + 1 & \text{if } i \equiv 0 \pmod{3} \text{ and } i \equiv 0 \pmod{2}. \end{cases} \quad (2)$$

Lemma 6. *The functions f defined by Eqs. (1) and (2) is a k -incidence ranking of P_n , where $k = \lceil \log_2 \frac{2n-1}{3} \rceil + 2$.*

Proof. Obviously, for $n \leq 10$, we can check the labels of P_n in Figure 1 to assure that f is an incidence ranking. Let $n = 3h + \ell > 10$, where $\ell \in \{0, 1, 2\}$. According to f , we observe that incidences (v_{3i}, e_{3i}) for $i = 1, 2, \dots, h$ are assigned to a larger label. In particular, if i is odd, then $f(v_{3i}, e_{3i}) = 4$. Otherwise, we consider a label $j \geq 5$. Let (v_{3i}, e_{3i}) be an incidence with label $f(v_{3i}, e_{3i}) = j$ and assume that it is on the rightmost of P_n as possible. Clearly, either j is a unique

label or there is another incidence $(v_{3i'}, e_{3i'})$ with $i' < i/2$ such that $f(v_{3i'}, e_{3i'}) = j$. If it is in the latter case, by Eq. (2), we have $f(v_{6i'}, e_{6i'}) = j + 1$, where the incidence $(v_{6i'}, e_{6i'})$ is between $(v_{3i'}, e_{3i'})$ and (v_{3i}, e_{3i}) in P . This shows that f is an incidence ranking.

We now calculate the largest label of the incidence ranking of P_n . If $n \leq 6$, we can determine the largest label of the incidence ranking in Figure 1. If $n \geq 7$, for convenience, we write $F(i)$ instead of $f(v_i, e_i)$ in Eq. (2). Thus, the largest label of the incidence ranking will occur in the last iteration of the recurrence $F(i) = F(i/2) + 1$ for $i = 6 \times 2^t$ and $t = 0, 1, \dots, m$. Obviously, if $t = 0$, we have the initial condition $F(6) = F(3) + 1 = 5$. Since $6 \times 2^m \leq n-1$, it follows that $m \leq \log_2 \frac{n-1}{6}$. We now solve the recurrence to obtain $F(6 \times 2^m) = F(6) + m \leq 5 + \log_2 \frac{n-1}{6}$. Since k is an integer and $n \geq 7$, we have $k = F(6 \times 2^m) = \lfloor \log_2 \frac{n-1}{6} \rfloor + 5 = \lfloor \log_2 \frac{2n-1}{3} \rfloor + 2$ through a reformulation. $\square$

Theorem 7. $\chi_{ir}(P_n) = \lfloor \log_2 \frac{2n-1}{3} \rfloor + 2$ for $n \geq 2$.

Proof. From Lemma 6, we obtain an upper bound of $\chi_{ir}(P_n)$. To establish a lower bound, we consider a binary tree structure which can be used to represent an incidence ranking of a path. Let f be an incidence ranking of P_n with the largest label k . We construct a binary tree T_f using the following rules: (i) for every incidence $i \in I(P_n)$, there is a corresponding node v_i in T_f ; (ii) for any two distinct incidences $i, j \in I(P_n)$, if i appears in the front of j on P , then v_i precedes v_j in the inorder traversal of T_f ; and (iii) if $f(i) = f(j)$, then there is a common ancestor of v_i and v_j in T_f , say $v_\ell (\neq v_i, v_j)$, such that $f(\ell) > f(i)$. For instance, Figure 2 shows two incidence rankings of P_6 and P_{10} , respectively, and their corresponding binary trees. It is clear that if both the left child and the right child of a node in

T_f are leaves, then their corresponding incidences in P_n are adjacent, and thus they cannot receive the same label. It is well-known that the number of leaves in a full binary tree with k levels is 2^{k-1} . Thus, if we avoid the case that both the children of a node are leaves, a binary tree with k levels has at most $(2^k - 1) - 2^{k-2} = 3 \times 2^{k-2} - 1$ nodes (e.g., the removal of all right children which are leaves in a full binary tree; see also Figure 2 for the tree with $k = 5$ that includes all light nodes.) Since P_n has $2n - 2$ incidences and all incidence must be mapped into nodes of a binary tree with level k , we have $2n - 2 \leq 3 \times 2^{k-2} - 1$. Since k is an integer, this implies that $k \geq \lceil \log_2 \frac{2n-1}{3} \rceil + 2$. $\square$

Let $C_n = v_0, e_1, v_1, e_2, v_2, \dots, e_{n-1}, v_{n-1}e_0v_0$ denote a cycle on n vertices. We now carry on the study of incidence ranking of C_n .

Theorem 8. $\chi_{ir}(C_n) = \lceil \log_2 \frac{2n-1}{3} \rceil + 3$ for $n \geq 3$.

Proof. For $n \geq 3$, let $k = \lceil \log_2 \frac{2n-1}{3} \rceil + 3$. To show that C_n admits a k -incidence ranking, a construction can be carried out by using the incidence ranking of P_n described in Lemma 6 along with a labeling of two additional incidences (v_0, e_0) and (v_{n-1}, e_0) . Since the labels of the first two incidences of P_n are $f(v_0, e_1) = 1$ and $f(v_1, e_1) = 2$, we assign the label k to the incidence (v_0, e_0) and consider the following three cases.

Case 1: $n \equiv 0 \pmod{3}$. In this case, the labelings of the last two incidences of P_n are $f(v_{n-1}, e_{n-1}) = 1$ and $f(v_{n-2}, e_{n-1}) = 3$. Thus, we can assign the label $f(v_{n-1}, e_0) = 2$.

Case 2: $n \equiv 1 \pmod{3}$. In this case, the labelings of the last two incidences of P_n are $f(v_{n-1}, e_{n-1}) = 4$ and $f(v_{n-2}, e_{n-1}) = 2$. Thus, we can assign the label $f(v_{n-1}, e_0) = 3$.

Case 3: $n \equiv 2 \pmod{3}$. In this case, the labelings of the last two incidences of P_n are $f(v_{n-1}, e_{n-1}) = 2$ and $f(v_{n-2}, e_{n-1}) = 1$. Thus, we can assign the label $f(v_{n-1}, e_0) = 3$.

Next, we consider the lower bound of $\chi_{ir}(C_n)$. Clearly, $\chi_{ir}(C_3) = 4$. In what follows, we assume that $n \geq 4$. Since C_n contains P_n as a subgraph, by Lemma 2, we have $\chi_{ir}(C_n) \geq \chi_{ir}(P_n) = k - 1$. Also, from Lemma 1, we know that there is an instance $(v_i, e_i) \in I(P_n)$ such that the label $f(v_i, e_i) = k - 1$ is unique in the incidence ranking of P_n . Let $Q = v_0, e_1, v_1, e_2, v_2, \dots, e_i, v_i$ be a subpath of P_n . Clearly, for each integer $j = 1, 2, \dots, k - 1$, there is an incidence $\ell \in I(Q)$ such that $f(\ell) = j$. Without loss of generality we assume that ℓ is in the forefront of Q as possible. By this assumption, every incidence before ℓ on Q must have a label less than j . Thus, the label of the incidence (v_0, e_0) cannot be assigned to j . Since j is arbitrary choose from $\{1, 2, \dots, k - 1\}$, it follows that the label of the incidence (v_0, e_0) must be assigned to k . $\square$

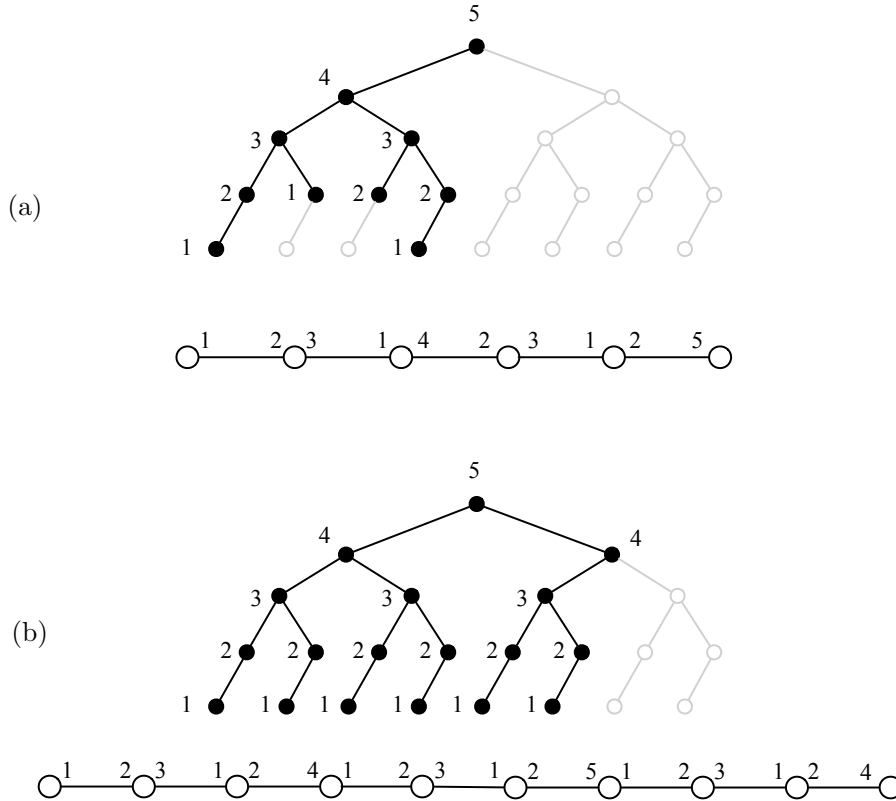


Figure 2: Two incidence rankings of P_6 and P_{10} and their corresponding binary trees.

4 $\chi_{ir}(G_{m,n})$ for $m = 2, 3$

In this section, we investigate the incidence ranking problem on grid graphs $G_{m,n}$. For $m = 2, 3$ and $n \geq 2$ arbitrary, we acquire the numbers of $\chi_{ir}(G_{m,n})$. The strategy in each case will be to establish a lower bound by Lemma 5 and then construct a k -incidence ranking where k equals that lower bound. By contrast, For $m = 4, 5$ and $n \geq 2$ arbitrary, we obtain upper bounds of $\chi_{ir}(G_{m,n})$.

Theorem 9. $\chi_{ir}(G_{2,n}) = \lceil \log_2 n \rceil \times 2 + 3$

Proof. Let $h = \lceil \log_2 n \rceil$. We first establish the lower bound of $\chi_{ir}(G_{2,n})$. By Lemma 5, we need to find a minimal edge-cut S of $G_{2,n}$ such that $\chi_{ir}^*(G_{2,n}|S)$ is minimum. Since $\chi_{ir}^*(G_{2,n}|S) = |S| + \chi_{ir}(G_{2,n} - S)$ and $G_{2,n} - S$ contains exactly two components, the best way to choose a minimal edge-cut of $G_{2,n}$ is that it contains exactly two edges. This shows that $\chi_{ir}(G_{2,n}) \geq \chi_{ir}(G_{2,\lceil n/2 \rceil}) + 2$. Since $G_{2,2}$ is isomorphic to a 4-cycle, by Theorem 8, we have $\chi_{ir}(G_{2,2}) = 5$. Solving the above recurrence, we obtain $\chi_{ir}(G_{2,n}) \geq \sum_{i=1}^{h-1} 2 + \chi_{ir}(G_{2,2}) = 2h + 3$.

Next, we prove $\chi_{ir}(G_{2,n}) \leq 2h + 3$ by induction on h (i.e., we provide a recursive construction to obtain a $(2h + 3)$ -incidence ranking of $G_{2,n}$). For the base case, we have $\chi_{ir}(G_{2,2}) = 5$ by Theorem 8. For example, Figure 3(a) shows two optimal 5-incidence rankings of $G_{2,2}$. For convenience, such two prescribed incidence rankings are called type A and type B incidence rankings, respectively. Let $h \geq 1$ be any integer. We suppose $\chi_{ir}(G_{2,2^h}) \leq 2h + 3$ and there is a prescribed $(2h + 3)$ -incidence ranking of $G_{2,2^h}$.

We now consider an incidence ranking of $G_{2,2^{h+1}}$ as follows. By induction hypothesis, we employ two

disjoint copies of $G_{2,2^h}$ with prescribed incidence rankings and join them with a pair of edges, called upper edge and lower edge, to form a $G_{2,2^{h+1}}$. If $h = 1$, one copy of $G_{2,2}$ is prescribed to type A incidence ranking and the other is prescribed to type B incidence ranking. In addition, two incidences on the upper edge (respectively, lower edge) are labeled by 6 and 1 (respectively, 7 and 2). See Figure 3(b) for the prescribed 7-incidence ranking of $G_{2,4}$. If $h \geq 2$, the two copies of $G_{2,2^h}$ are prescribed to the same incidence ranking. Also, two incidences on the upper edge (respectively, lower edge) are labeled by $2h + 4$ and 2 (respectively, $2h + 5$ and 1). Thus, we obtain a $(2h + 5)$ -incidence ranking of $G_{2,2^{h+1}}$ as the prescribed incidence ranking. It follows that $\chi_{ir}(G_{2,2^{h+1}}) \leq 2(h + 1) + 3$. Note that, for $r \in \{2^h + 1, 2^h + 2, \dots, 2^{h+1} - 1\}$, $G_{2,r}$ is a subgraph of $G_{2,2^{h+1}}$. By Lemma 2, we have $\chi_{ir}(G_{2,r}) \leq \chi_{ir}(G_{2,2^{h+1}}) = 2(h + 1) + 3 = 2 \lceil \log_2 r \rceil + 3$. For instance, Figure 3(c) shows an optimal incidence ranking of $G_{2,10}$.

Consequently, combining the lower bound and the recursive construction yields the desired result. $\square$

Theorem 10. $\chi_{ir}(G_{3,n}) = \lceil \log_2 n \rceil \times 3 + 4$

Proof. Let $h = \lceil \log_2 n \rceil$. For the lower bound, by Lemma 5, the best way to choose a minimal edge-cut of $G_{3,n}$ is that it contains exactly three edges. Thus, a proof similar to Theorem 9 shows that $\chi_{ir}(G_{3,n}) \geq \chi_{ir}(G_{3,\lceil n/2 \rceil}) + 3$. Again, by Theorem 9, we know $\chi_{ir}(G_{3,2}) = \chi_{ir}(G_{2,3}) = 7$. Solving the above recurrence, we obtain $\chi_{ir}(G_{3,n}) \geq \sum_{i=1}^{h-1} 3 + \chi_{ir}(G_{3,2}) = 3h + 4$.

Similarly, we can use induction (i.e., a recursive construction) to prove $\chi_{ir}(G_{3,n}) \leq 3h + 4$, and obtain a $(3h + 4)$ -incidence ranking of $G_{3,n}$. The base

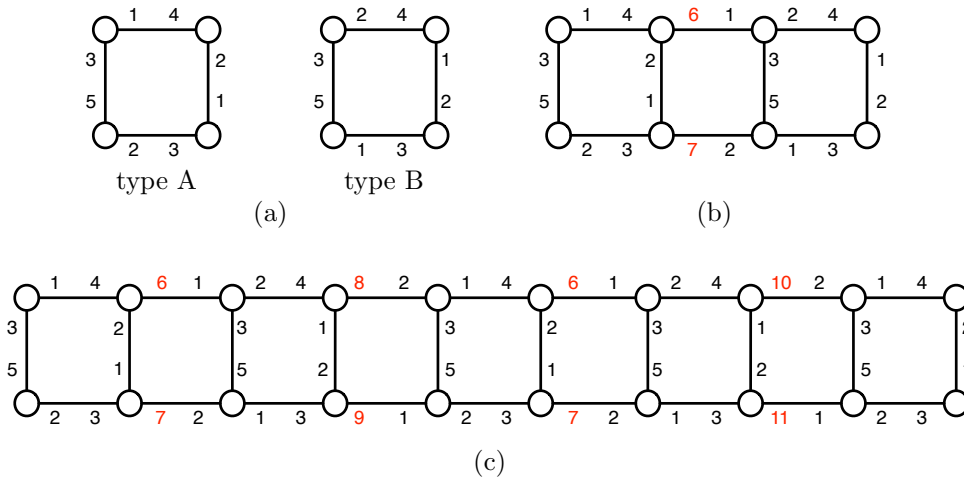


Figure 3: Optimal incidence rankings of $G_{2,n}$: (a) $n = 2$ (type A and type B); (b) $n = 4$; and (c) $n = 10$.

case is the graph $G_{3,2}$ and $\chi_{ir}(G_{3,2}) = 7$. There are two prescribed incidence rankings of $G_{3,2}$, called type A and type B incidence rankings (see the graphs encircled by the dashed cycles in Figure 4). Then, using a construction similar to the proof of Theorem 9 that adds three extra edges for each iteration, we can construct the desired incidence rankings. For instance, Figure 4 shows an optimal incidence ranking of $G_{3,10}$. $\square$

Theorems 11 and 12 provide upper bounds of

$\chi_{ir}(G_{m,n})$ for $m = 4, 5$, respectively. Since all the proofs are similar to the constructions of Theorem 9 and Theorem 10, we omit details of the proofs. According to the conditions of theorems, we check that $\lceil \log_2(8/2) \rceil = \lceil \log_2(8/3) \rceil = 2$ and $\lceil \log_2(9/2) \rceil = 3 > \lceil \log_2(9/3) \rceil = 2$. Figure 5 and Figure 6 show the optimal incidence rankings of $G_{4,8}$ and $G_{4,9}$, respectively. Figure 7 and Figure 8 show the optimal incidence rankings of $G_{5,8}$ and $G_{5,9}$, respectively.

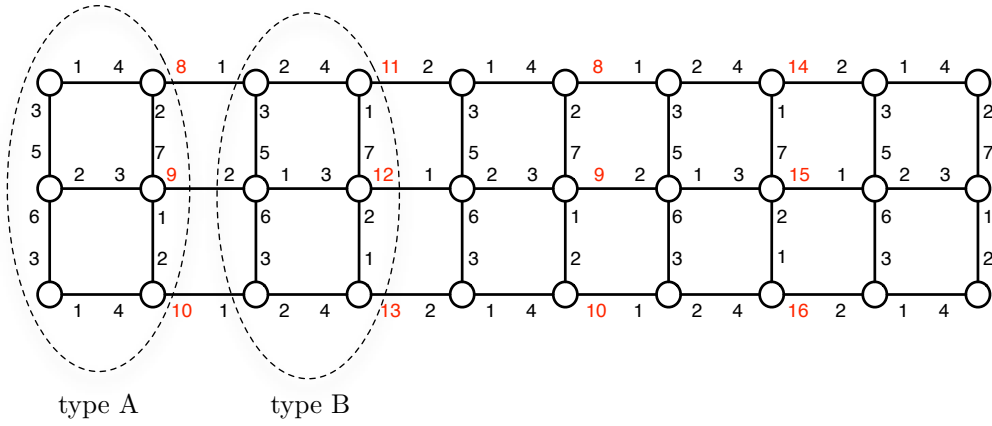


Figure 4: An optimal incidence ranking of $G_{3,10}$

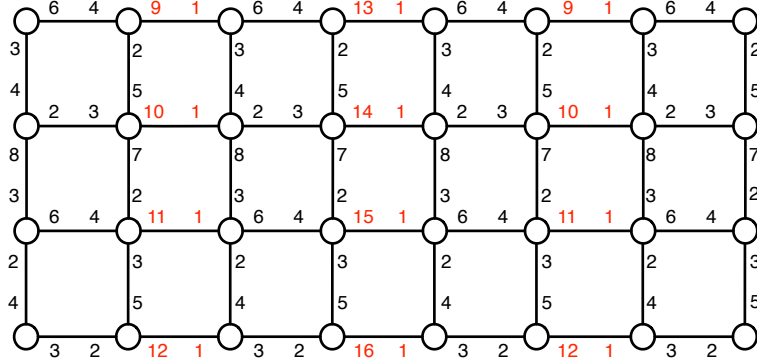


Figure 5: An optimal incidence ranking of $G_{4,8}$

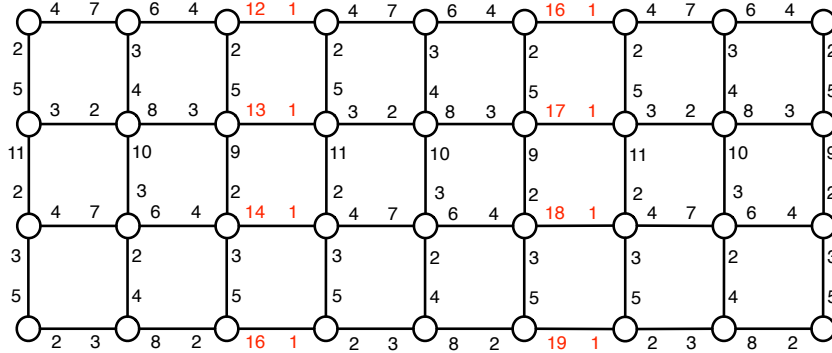
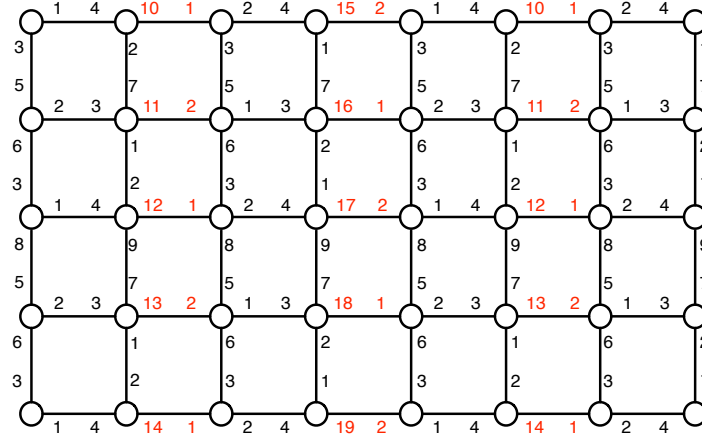
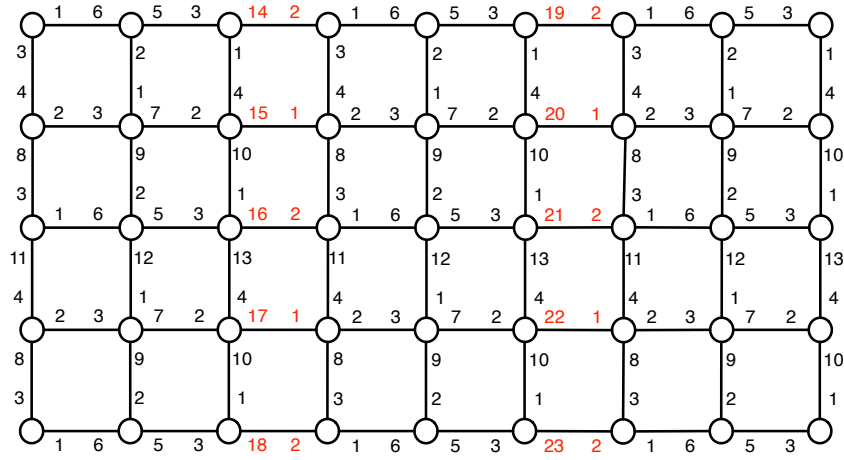


Figure 6: An optimal incidence ranking of $G_{4,9}$

Figure 7: An optimal incidence ranking of $G_{5,8}$ Figure 8: An optimal incidence ranking of $G_{5,9}$

Lemma 11.

$$\chi_{ir}(G_{4,n}) \leq \begin{cases} \lceil \log_2 n \rceil \times 4 + 4, & \text{if } \lceil \log_2(n/2) \rceil = \lceil \log_2(n/3) \rceil \\ \lceil \log_2 \lceil n/3 \rceil \rceil \times 4 + 11, & \text{if } \lceil \log_2(n/2) \rceil \neq \lceil \log_2(n/3) \rceil \end{cases}$$

Lemma 12.

$$\chi_{ir}(G_{5,n}) \leq \begin{cases} \lceil \log_2 n \rceil \times 5 + 4, & \text{if } \lceil \log_2(n/2) \rceil = \lceil \log_2(n/3) \rceil \\ \lceil \log_2 \lceil n/3 \rceil \rceil \times 5 + 13, & \text{if } \lceil \log_2(n/2) \rceil \neq \lceil \log_2(n/3) \rceil \end{cases}$$

5 Concluding remarks

In this paper, we newly define the incidence ranking problem of graphs. As a consequence, we obtain the incidence ranking numbers $\chi_{ir}(P_n)$, $\chi_{ir}(C_n)$, and $\chi_{ir}(G_{m,n})$ for $m = 2, 3$ and $n \geq 2$ arbitrary. Moreover, upper bounds of $\chi_{ir}(G_{m,n})$ for $m = 4, 5$ are also acquired.

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